



Teheran University of Medical Sciences

International campus Vice-Dean for Educational Affairs

No. 124, Baradaran-e-Mozafar St.
Dameshgh St., Vali-e-Asr Ave.
Tehran 3439123900 Iran.
Tel: (+98) 21-88 89 36 78
Web: gsia.tums.ac.ir/en/icedu
Email: icedu@tums.ac.ir

Final Exam
International Campus
Cell & Molecules, MD Program
September 2023 semester

Name and Surname:

Student ID Number:

Course Coordinator: Dr. Panahi

Date of Exam: Jan. 22, 2024

Duration of Exam: 87 minutes

MCQ Question

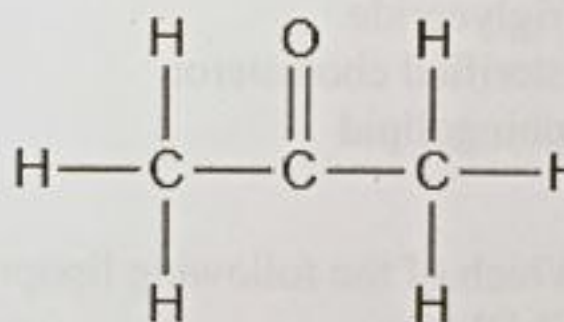
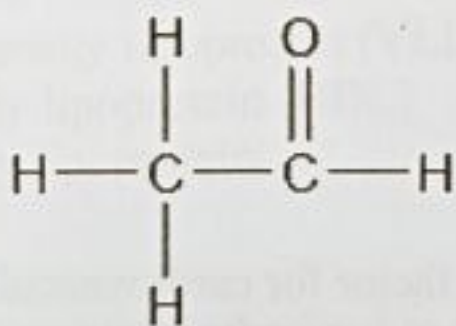
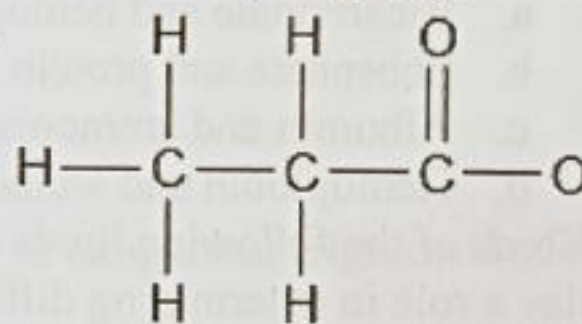
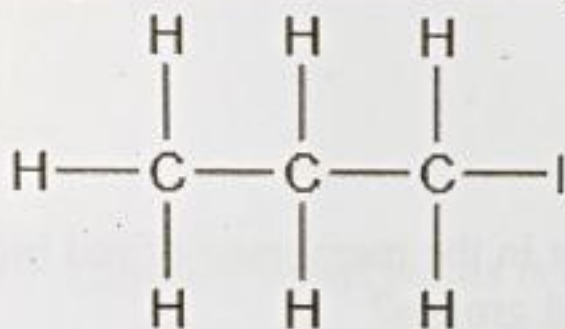
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Directions:

- Each of the following questions contains one correct answer only.
- Please choose the correct answer for the MCQs in the answer sheet with pencil.

Biochemistry: Dr. Panahi

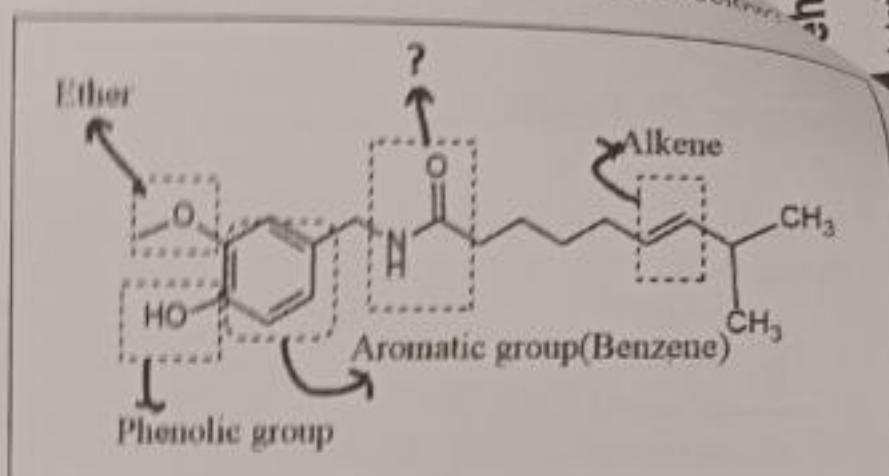
- Some biologically-important molecules dissolve easily in water because the water molecule is
 - polar
 - organic
 - saturated
 - a polymer
- The maintenance of a constant pH of the blood is achieved by ...
 - acids
 - bases
 - water
 - buffers
- Which of the following is true of hydrogen bonds between water molecules?
 - form because water molecules contain non-polar covalent bonds
 - they hold together the oxygen and the hydrogen atoms of each molecule
 - attract the hydrogen end of one molecule to the oxygen end of another molecule
 - they form between positively and negatively charged ions in solution
- Identify the compound which has *carboxylic acid* as the functional group.





5. What is the name of the group shown as question mark in the structure below?

- a. Carboxylic acid
- b. Halide
- c. Amide
- d. Ketone



6. Aspirin is an acid with a $pK_a = 3.4$. At the blood $pH = 7.4$, what would be the ratio of A^- to HA ?
- a. 4
 - b. 4000
 - c. 10000
 - d. 1/10000
7. Which of the following will be best to make a buffer of $pH = 7.20$?
- a. CH_3COOH and CH_3COOK , $pK_a = 4.752$
 - b. $HClO_2$ and $KClO_2$, $pK_a = 1.96$
 - c. NH_3 and NH_4Cl , $pK_a = 9.2$
 - d. $HClO$ and $KClO$, $pK_a = 7.54$
 - e.
8. The primary buffer systems in the cytoplasm (Intracellular fluid)?
- a. Bicarbonate and hemoglobin
 - b. Phosphate and protein
 - c. Albumin and ammonia
 - d. Hemoglobin and ammonia
9. Which of the following lipids are present in the membrane of red blood cells (RBC) and play a role in determining different blood groups?
- A- Phospholipid
 - B- Triglyceride
 - C. Esterified cholesterol
 - D- sphingolipid
10. Which of the following lipoproteins is a risk factor for cardiovascular diseases?
- A- VLDL
 - B- Chylomicron
 - C- HDL
 - D- LDL

11. Which of the following drugs inhibits phospholipase A2 (PLase A2) in the synthesis of prostaglandins?
- A- Aspirin
 - B- Celecoxib
 - C- Corticosteroids
 - D- Indomethacin
12. The body is unable to synthesize which of the following fatty acids and should be provided by the diet?
- A- Linoleic acid
 - B- Stearic acid
 - C- Palmitic acid
 - D- oleic acid
13. Which of the following options is correct regarding the dominant apoprotein (Apo) in each lipoprotein?
- A- Apo -A---> Chylomicron
 - B- Apo-B48---> HDL
 - C- Apo-B100---> LDL
 - D- Apo-B48---> VLDL
14. Free cholesterol (FC) is the precursor of all the following compounds except:
- A- Vitamin D (Vit D)
 - B- Bile acids
 - C- Vitamin A (Vit A)
 - D- Sex hormones
15. Which of the following lipoproteins is the carrier of exogenous triglycerides or TGexo diet?
- A- Chylomicron (Chy)
 - B- Very low-density lipoprotein (VLDL)
 - C- High-density lipoprotein (HDL)
 - D- Low density lipoprotein
16. Prostaglandins are synthesized in the body from
- a) Myristic acid
 - b) Arachidonic acid
 - c) Stearic acid
 - d) Lignoceric acid

d) Can be present anywhere in proteins

$\text{pK}_{\text{a NH}_2} = 9.3$

- d) 4.25

d) Cysteine

d) Arginine

d) Increase of temperature

d) Present in heart and skeletal muscle

23. Name the protein which is responsible for extensibility and elasticity of connective tissues?
- a) Collagen
 - b) Elastin
 - c) Actin
 - d) Keratin
24. Which of the following correctly describes the Bohr effect?
- a) Effect of temperature on the binding and release of oxygen by hemoglobin
 - b) Effect of 2,3-BPG on the binding and release of oxygen by hemoglobin
 - c) Effect of Carbon monoxide on the binding and release of oxygen by hemoglobin
 - d) Effect of pH and CO_2 concentration on the binding and release of oxygen by hemoglobin
25. Which of the following reaction is catalyzed by isomerase?
- a) Hydrolysis reaction
 - b) Oxidation-Reduction reaction
 - c) Intramolecular rearrangement of bonds
 - d) Transfer of group from one molecule to another
26. According to the Michaelis-Menten Kinetics, when substrate concentration is very high:
- a) $V_0 = 2 V_{\max}$
 - b) $V_0 = V_{\max}/2$
 - c) $V_0 = V_{\max}$
 - d) $V_0 = V_{\max}/4$
27. Which of the following statement is not true about the K_m (Michaelis constant)?
- a) Has a lower numerical value when the enzyme has high affinity for its substrate.
 - b) Is the substrate concentration when the reaction rate is half of the maximum rate.
 - c) Is a measure of the affinity of the enzyme for its substrate.
 - d) Has a higher numerical value when the enzyme has high affinity for its substrate.



28. Enzymes synthesized in inactive form are called:

- a) Proenzymes
- b) Coenzymes
- c) Holoenzymes
- d) Cofactors

29. Which statement is incorrect about allosteric enzymes?

- a) Display a sigmoidal dependence on substrate concentration
- b) Frequently catalyze the committed step early in a pathway
- c) Are almost always composed of multiple subunits
- d) An allosteric substrate functions as a negative effector

30. Inhibition of HMG-CoA reductase by Atorvastatin falls into which category of inhibition?

- a) Competitive inhibition
- b) Irreversible inhibition
- c) Substrate inhibition
- d) Non-competitive inhibition

31. Which of the following statement about isoenzymes is not true:

- a) Display different regulatory properties
- b) Have distinct physical properties
- c) Catalyze the different reaction
- d) May be expressed in a tissue-specific

32. Succinate is the substrate for succinate dehydrogenase that converts succinate to fumarate. In the presence of reversible competitive inhibitor like malonate in place of succinate, the enzymes:

- a) K_m increases and V_{max} remains the same
- b) Both K_m and V_{max} increases
- c) Both K_m and V_{max} decreases
- d) K_m decreases and V_{max} remains the same

33. Which of the following monosaccharides is the precursor for synthesis of glycogen as a storage carbohydrate in animals?

- a) Fructose
- b) Glucose
- c) Ribose
- d) Mannose

34. α -glucose and β -glucose are:

- a) Stereoisomer
- b) Epimers
- c) Anomers
- d) Keto-Aldo pairs

35. Which of the following are enantiomers:

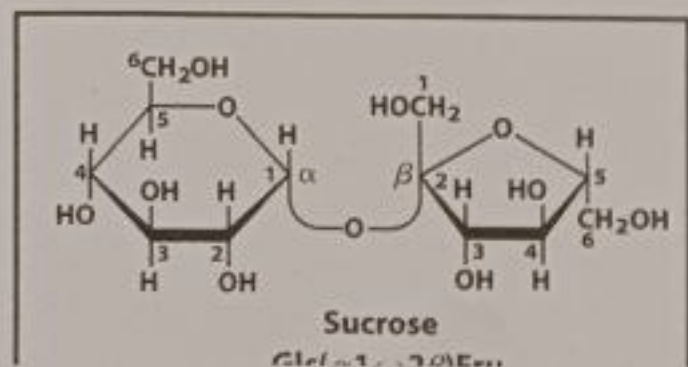
- a) D-Mannose and L-Glucose
- b) d- Galactose and I-Glucose
- c) D-Galactose and L-Galactose
- d) d-Mannose and I-Mannose

36. Which of the following monosaccharides can act as a structural component for biosynthesis of ATP?

- a) Mannose
- b) Ribose
- c) Arabinose
- d) Ribulose

37. The following structure is related to sucrose. Which statement is true about it?

- a) Reducing disaccharides
- b) Nutritionally important disaccharide
- c) It is composed of two aldohexose
- d) It can be found mostly in milk





38. Which statement is true about amylose and amylopectin?
- a) Both of them contain alpha linkages
 - b) Amylose contain alpha linkage but amylopectin contains beta linkage
 - c) Amylopectin has a linear structure but amylose contains branched chain
 - d) Both of them contain branched chain
39. One dermatologists want to design a substance that is used to prevent and treat blood clots in blood vessels and the heart. Which of the following carbohydrates will be suggested?
- a) Heparin
 - b) Keratan sulphate
 - c) Chondrotin sulphate
 - d) Heparan sulphate
40. Which of the following molecules cannot be hydrolyzed with human intestinal enzymes?
- a) Cellulose
 - b) Maltose
 - c) Lactose
 - d) Sucrose
41. Which enzyme is responsible for unwinding the DNA double helix during replication?
- a. DNA polymerase
 - b. Helicase
 - c. Ligase
 - d. RNA polymerase

42. Which of the following are involved in the termination of replication?

A) DNA polymerase

B) Helicase

c) SSB

D) Tus protein

43. All of the following are inhibitors of bacterial DNA gyrase except:

A) Ciprofloxacin

b) Doxorubicin

C) Nalidixic acid

d) Novobiocin

44. What is the function of DNA ligase in DNA replication?

a. Unwinding the DNA helix

b. Connecting Okazaki fragments

c. Synthesizing RNA primers

d. Proofreading DNA strands

45. The two strands of DNA are held together by:

a. Covalent bonds

b. Ionic bonds

c. Hydrogen bonds

d. Van der Waals forces



46. Which of the following mechanisms is used to repair a double-strand when a similar sequence is not available?

- a) Homologous genetic recombination
- b) Site specific recombination
- c) Replicative recombination
- d) Non-homologous recombination

47. Which of the following enzymes initiates the Excision Repair Base repair process?

- a) Uvr ABC excinuclease
- b) Deoxyribose phosphate lyase
- c) Dam Methylase
- d) DNA uracil glycosylase

48. Sickle cell anemia is caused by which of the following mutations?

- A- Missense
- B- Nonsense
- C- Silent
- D- Deletion

49. Which of the following is required for the transfer of the newly formed Peptidyl-tRNA from the A site to the P site by the translocase?

- A- EF-G, GTP
- B- EF-TU, GTP
- C- Peptidyl transferase, GTP
- D- EF-TS, GDP

50. The binding of prokaryotic RNA polymerase to the promoter of genes is inhibited by which of the following antibiotics?
- A- Puromycin
 - B- Terramycin
 - C- Rifamycin
 - D- Streptomycin
51. All of the following are present in the structure of polysomes, except:
- A- DNA
 - B- rRNA
 - C- mRNA
 - D- tRNA
52. Which antibiotic inhibits bacterial peptidyl transferase enzyme?
- A- Tetracycline
 - B- Gentamicin
 - C- Chloramphenicol
 - D- Puromycin
53. Which of the following has a major role in selecting the correct amino acid during protein synthesis?
- A- tRNA-UAG
 - B- mRNA
 - C- P site of ribosome
 - D- amino acid tRNA synthetase
54. Which of the following are inducers and repressors of gene expression?
- A- CAP elements
 - B- Trans-acting factors
 - C- Cis-acting elements
 - D- Ligand proteins
55. How does cAMP regulate the lactose operon?
- A- binding to lactose suppressor
 - B- binding to CAP and inhibiting the promoter
 - C- binding to the operator and starting transcription
 - D- binding to CAP and increasing transcription



56. RNA polymerase holoenzyme is attached to the lactose operon at which of the following location?

- A- gene i
- B- gene z
- C- operator
- D- Promoter

57. The vitamin required for the formation of hydroxyproline (in collagen) is

- (A) Vitamin E
- (B) Vitamin A
- (C) Vitamin D
- (D) Vitamin C

58. Which of the following **is not** a function of Vitamin D?

- a) Regulation of calcium and phosphorus levels in blood
- b) Hydroxylation of proteins of connective tissue
- c) Normal insulin secretion by pancreas
- d) Bone mineralization

59. Which vitamin is a methyl group acceptor?

- a) Coenzyme A
- b) Lipoic acid
- c) Vit B12
- d) Biotin

60. Which of the following vitamins provide the coenzyme for γ -carboxylation reactions?

- a) Biotin
- b) Menaquinone
- c) Pyridoxine
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61. Vitamin-E deficiency can cause
- a) Rickets
 - b) Anemia
 - c) Pellagra
 - d) Beriberi
62. Which of the following nutrient deficiency causes megaloblastic anaemia?
- a) Cobalamin
 - b) Niacin
 - c) Folic acid
 - d) Retinol
63. Which of the following is the active form of pantothenic acid?
- a) THF
 - b) Coenzyme A
 - c) PLP
 - d) TPP
64. Which of the following organs are responsible for the activation of vitamin D in the human body?
- a) Intestine and Liver
 - b) Kidney and Pancreas
 - c) Liver and Kidney
 - d) Stomach and Brain

Histology: Dr.Rastegar

- 65- In receptor-mediated endocytosis
- a) Clathrin protein is dispersed in the cell membrane.
 - b) Binding of the ligand to the receptor causes the accumulation of clathrin in certain areas.
 - c) Binding of the ligand to the receptor causes the accumulation of RER close the membrane.
 - d) The vesicle merges with the proteasome of the cell.



66- Which of this structure DONOT enclose by membrane?

- a) Proteasome b) Peroxisome c) Lysozyme d) Endosome

67- In the process of protein synthesis in the cell, the signal recognition particle

- a) Was seen in the RER membrane.
b) Facilitates binding of signal peptide receptor to ribosome.
c) Prevents protein elongation in the cytoplasm
d) Causes the transporter of the translocator complex.

68- Protein synthesis for export does in:

- a. Ribosomes b. Golgi apparatus c. SER d. RER

69- Which of the following is CORRECT about the organelles inside the cell?

- a) Transport vesicles of the Golgi apparatus move along microtubules.
b) Purin protein is present in the inner membrane of mitochondria.
c) Denatured proteins merge with the endosomal part of the cell.
d) Vimentin is found in the cytoplasm around the mitochondria.

70- In the structure of the cell nucleus

- a) Nucleosome contains histone and DNA.
b) Nucleosome contains histone and chromatin.
c) DNA is transcribed in the compact structure of euchromatin.
d) The main structure of nucleosome core consists of four histones.

71 -Which one of these organelles can replicate before the cell division stage?

- a -RER b- mitochondria c-SER d-Ribosome

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72- Pale, large and centrally located nucleus in the cell, shows:

- a. This cell is dead
- b. This cell starts apoptosis
- c. This cell will differentiate
- d. This cell is active in protein synthesis

73- In which of the following cases the protein not related to the relevant organelle?

- a. Proteasome – Ubiquitin
- b. Peroxisome-catalase
- c. RER - Clathrin
- d. Mitochondria-Pu

Physiology: Dr.Ranjbaran

Assume that the concentration of ion x in the extracellular fluid is 300 mEq/lit and inside the cell is 20 mEq/lit, the resting membrane potential is closer to $\log_{10}(15) = 1.17$ $\log_{10}(0.06) = -1.2$		74
+71 mv	الف	
-73 mv	ب	
+94 mv	ج	
-61 mv	د	
Comparing Goldman equation with Nernst equation, which of the following is correct?		75
The permeability of the membrane to ions is only considered in the Nernst equation.	الف	
The Nernst equation shows the simultaneous effect of several ions on the membrane potential.	ب	
In the Nernst equation, only the concentration gradient of positively charged ions is included in the formula.	ج	
Goldman equation describes real conditions, while Nernst equation depicts imaginary conditions.	د	
Which of the following <u>does not play</u> any role in determining the level of resting membrane potential?		76
Leakage of K through the K leak channels	الف	
Leakage of Na through the K leak channels	ب	
Leakage of Na through the Na leak channels	ج	
Na-K Pump	د	



The peak of an action potential in a large nerve cell is closer to the Nernst potential of which of the following ions?		
K	الف	
Na	ب	
Cl	ج	
HCO ₃	د	
منبع:		
What is the main cause of repolarization phase of an action potential?		
		78
Closure of voltage gated K channels	الف	
Opening of voltage gated K channels	ب	
Opening of voltage gated Ca channels	ج	
Closure of voltage gated Na channels	د	
A new AP cannot occur in an excitable fiber as long as the membrane is still depolarized from the preceding action potential. Why?		
		79
Because of inactivation of voltage gated Na channels	الف	
Because of inactivation of voltage gated K channels	ب	
Because of activation of voltage gated Na channels	ج	
Because of activation of voltage gated K channels	د	
Salutatory conduction is of value for		80
decrease in conduction velocity	الف	
conserving energy	ب	
having plateaue phase	ج	
existence of schwann cells	د	
In plateaue phase, calcium is transported inside the cardiac cell by		81
Na ⁺ /Ca ²⁺ -exchanger	الف	
Ca ²⁺ -ATPase	ب	
Na ⁺ /K ⁺ -ATPase pump	ج	
voltage gated ca channels	د	
منبع:		

which compartment is the largest amount of body water found?		8
ECF		2
ICF	الف	
Plasma	ب	
Interstitial fluid	ج	
Which of the following acts as β cell glucose sensor?	د	
		83
GLUT 1		
GLUT 2	الف	
GLUT 3	ب	
GLUT 4	ج	
Which of the following transporters has the most number in the kidneys?	د	
		84
SGLT 1		
SGLT 2	الف	
SGLT 3	ب	
SGLT 4	ج	
Simple diffusion of ions is done based on:	د	
		85
chemical gradient		
concentration gradient	الف	
electrical gradient	ب	
electrochemical gradient	ج	
Which of the following is negatively correlated with the rate of simple diffusion?	د	
		86
Lipid solubility	الف	
Thickness of membrane	ب	
Concentration gradient across membrane	ج	
Surface area of membrane	د	
Which of the following is the most important factor limiting the lateral movement of proteins in the membrane of epithelial cells in the kidney?		87
Connecting proteins to the cell cortex	الف	
Connecting proteins to the extracellular matrix	ب	
Connecting the proteins of one cell to the proteins of the adjacent cell	ج	
The existence of tight junctions between cells	د	
		منبع:

فرمت پاسخ نامه سلول و ملکول

1		201		151		101	1	51	1	1
2		202		152		102	3	52	4	2
3		203		153		103	4	53	3	3
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		232		182		132	2	82	1	32
		233		183		133	2	83	2	33
		234		184		134	2	84	3	34
		235		185		135	4	85	3	35
		236		186		136	2	86	2	36
		237		187		137	4	87	2	37
		238		188		138		88	1	38
		239		189		139		89	1	39
		240		190		140		90	1	40
		241		191		141		91	2	41
		242		192		142		92	4	42
		243		193		143		93	2	43
		244		194		144		94	2	44
		245		195		145		95	3	45
		246		196		146		96	4	46
		247		197		147		97	4	47
		248		198		148		98	1	48
		249		199		149		99	1	49
		250		200		150		100	3	50